

LOCALIZED BOST–CONNES SYSTEMS II: IDEAL STRUCTURE, BOUNDARY QUOTIENT, AND INDEX

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ABSTRACT. We study the ideal structure, the boundary quotient, the K -theory and the index theory of the localized Bost–Connes system $\mathcal{A}_{K,S} = C(Y_S) \rtimes J_S$, and show that all of it is governed by one structural feature: since J_S is free abelian on S , each prime carries exactly *one* isometry and the range projections $\mathbf{1}_{aY_S}$ are *nested* rather than orthogonal.

From this we obtain: the groupoid is topologically principal, its isotropy on the finite closed orbit $\partial Y_S \cong \text{Cl}_S^+$ is $I_S^0 = \ker(I_S \rightarrow \text{Cl}_S^+)$, and off ∂Y_S it is trivial whenever K has finitely many totally positive units; the ideal lattice is then the lattice of open invariant subsets of $\tilde{Y}_S \setminus \partial Y_S$ glued to the lattice of closed subsets of the torus $\widehat{I_S^0}$, the latter contributing ideals that neither the diagonal nor the gauge action sees. The tight quotient makes the isometries unitary rather than Cuntz, giving $\partial \mathcal{A}_{K,S} \cong M_{h_S^+}(C^*(I_S^0))$, a type I algebra with a faithful trace whose only isomorphism invariant is the narrow class number h_S^+ , recoverable as the divisibility of $[1]$ in $K_0 = \Lambda^{\text{ev}}(I_S^0)$. For a single prime, $K_1(\mathcal{A}_{K,S}) = 0$ and $K_0(\mathcal{A}_{K,S}) \cong C(G_S, \mathbb{Z})/\mathbb{Z} \oplus \mathbb{Z}$ is free; for several primes K_* is the abutment of a collapsing Koszul spectral sequence which we do not evaluate.

The same feature governs the index theory. At the symmetric KMS_β state, every prime inclusion $M_{S_1} \subset M_{S_2}$ has infinite index for *every* conditional expectation, for the elementary reason that the valuation projections $\mathbf{1}_{\{v_p=k\}}$ lie in the relative commutant and are moved onto one another by partial isometries commuting with M_{S_1} ; there is no index formula in Np and $[\text{Cl}_{S_2}^+ : \text{Cl}_{S_1}^+]$, and no Galois correspondence with intermediate prime sets, there being a continuum of intermediate algebras where there are two intermediate prime sets. The finite index instead comes from symmetry breaking: $M_\varepsilon \cong M_{\text{sym}}^{\widehat{\Xi}_\beta} \subset M_{\text{sym}}$ is the trivial inclusion of a factor tensored with $\mathbb{C} \subset \ell^\infty(\widehat{\Xi}_\beta)$, of index $|\Xi_\beta|$, its intermediate algebras are indexed by the partitions of $\widehat{\Xi}_\beta$, and $\log[M_{\text{sym}} : M_\varepsilon] = |A_\beta| \log 2$ is also the relative entropy of an extremal phase against the symmetric one.

Finally we run the control experiment. Restoring the additive part gives $\mathcal{A}_{K,S}^{\text{aff}} = C(\widehat{\mathcal{O}}_S) \rtimes (\mathcal{O}_K \rtimes J_S)$, whose additive cosets provide *finite covers* and hence genuine Cuntz relations; it is a simple purely infinite Kirchberg algebra. Nested versus partitioning range projections is therefore exactly the dividing line. Even there, however, $[1]$ carries almost nothing: the Cuntz relation forces its order to divide $\gcd_{p \in S}(\text{Np} - 1)$, which is 1 as soon as S contains a prime of norm 2.

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1. THE SINGLE-ISOMETRY PHENOMENON

We keep the notation of [21]: K is a number field, S a set of finite primes, $J_S = \bigoplus_{\mathfrak{p} \in S} \mathbb{N}$ the free abelian monoid of integral ideals supported on S with enveloping group I_S , $Y_S = \widehat{\mathcal{O}}_S \times_{U_S} G_S$, and

$$\mathcal{A}_{K,S} = C(Y_S) \rtimes J_S = \overline{\text{span}}\{\mu_{\mathfrak{a}} f \mu_{\mathfrak{b}}^*\}, \quad \mu_{\mathfrak{a}}^* \mu_{\mathfrak{a}} = 1, \quad \mu_{\mathfrak{a}} \mu_{\mathfrak{a}}^* = \mathbf{1}_{\mathfrak{a}Y_S}.$$

The series. The foundational paper [21] classifies the KMS states of $\mathcal{A}_{K,S}$ and introduces the obstruction group Ξ_{β} . Five papers build on it, numbered I–V:

- I factor type and the spectrum of transitions
- II ideal structure, boundary quotient, and index (this paper)
- III invariants, entropy, and the measure class
- IV noncommutative metric geometry
- V the free variant and the free critical exponent

Paper I has appeared; Papers III–V are in preparation. *Nothing in this paper depends on them:* every statement below is proved here or cited from [21], and the forward references are signposts only. We refer to them by numeral, without citation, for that reason.

Recall the exact sequence [21, (2.1)]

$$(1.1) \quad 1 \rightarrow U_S / \overline{\mathcal{O}_{K,+}^{\times}} \rightarrow G_S \rightarrow \text{Cl}_S^+ \rightarrow 1, \quad h_S^+ := |\text{Cl}_S^+|.$$

Lemma 1.1 (Single isometry, nested projections). *J_S is free abelian on S . Hence:*

- (1) *for each $\mathfrak{p} \in S$ there is exactly one generating isometry $\mu_{\mathfrak{p}}$;*
- (2) *the range projections are nested: $\mathbf{1}_{\mathfrak{a}Y_S} \mathbf{1}_{\mathfrak{b}Y_S} = \mathbf{1}_{\text{lcm}(\mathfrak{a}, \mathfrak{b})Y_S}$;*
- (3) *there is no finite family of isometries in $\mathcal{A}_{K,S}$ with mutually orthogonal ranges summing to 1; indeed $Y_S \setminus \bigcup_{\mathfrak{p} \in S} \mathfrak{p}Y_S = \{[x, g] : x \in \widehat{\mathcal{O}}_S^{\times}\}$ is nonempty and clopen.*

Proof. Freeness is unique factorization of ideals. (2) is immediate from $\mathfrak{a}Y_S \cap \mathfrak{b}Y_S = \text{lcm}(\mathfrak{a}, \mathfrak{b})Y_S$. For (3), two of the range projections $\mathbf{1}_{\mathfrak{a}Y_S}$ are never orthogonal by (2); and if x is a unit at every $\mathfrak{p}$ then $[x, g] \notin \mathfrak{p}Y_S$ for any $\mathfrak{p}$, which gives the last assertion. That no family of isometries whatever has orthogonal ranges summing to 1 — $\mathcal{A}_{K,S}$ is not properly infinite — follows from Theorem 3.1 below: $\mathcal{A}_{K,S}$ maps unitally onto an algebra with a faithful tracial state τ , and n isometries with orthogonal ranges summing to 1 would give $n = \sum \tau(v_i^* v_i) = \sum \tau(v_i v_i^*) = 1$. $\square$

Lemma 1.1 is the source of every structural statement below. We flag its use by referring to it explicitly. The contrast is with the $ax + b$ algebras of Cuntz–Li and Cuntz–Deninger–Laca, where each ideal $\mathfrak{a}$ carries $N\mathfrak{a}$ isometries indexed by the additive cosets $a + \mathfrak{a}$, whose ranges *partition*; §6 restores that ingredient and confirms that it is the cause.

2. ISOTROPY, TOPOLOGICAL PRINCIPALITY, AND THE IDEAL LATTICE

Let $\mathcal{G}_S = I_S \times \widetilde{Y}_S$ be the dilated groupoid of [21, Prop. 2.8], with points $[x, g]$ subject to $(wx, g) \sim (x, r(w)g)$ for $w \in U_S$.

Lemma 2.1 (Isotropy formula). *For $y = [x, g]$ put $T(y) = \{\mathfrak{p} \in S : x_{\mathfrak{p}} = 0\}$. Then*

$$(2.1) \quad \text{Iso}_{\mathcal{G}_S}(y) = \{\mathfrak{a} \in I_{T(y)} : \sigma_{\mathfrak{a}} \in r(U_{T(y)})\}.$$

Proof. $\mathfrak{a} \cdot [x, g] = [x, g]$ means $\mathfrak{a}x = wx$ and $g = r(w)^{-1} \sigma_{\mathfrak{a}} g$ for some $w \in U_S$. Comparing $\mathfrak{p}$ -components: if $x_{\mathfrak{p}} \neq 0$ then $\mathfrak{p}^{a_{\mathfrak{p}}} x_{\mathfrak{p}} = w_{\mathfrak{p}} x_{\mathfrak{p}}$ forces $a_{\mathfrak{p}} = 0$ and $w_{\mathfrak{p}} = 1$; if $x_{\mathfrak{p}} = 0$ both sides vanish. So $\mathfrak{a} \in I_T$, $w \in U_T$, and the second equation is $\sigma_{\mathfrak{a}} = r(w)$. $\square$

Proposition 2.2 (Isotropy off the boundary). *Write $\mathbb{I}_T = \{u \in \mathbb{I}_S : u_{\mathfrak{p}'} = 1 \text{ for all } \mathfrak{p}' \in S \setminus T\}$. Then for y with $T(y) = T$,*

$$(2.2) \quad \text{Iso}_{\mathcal{G}_S}(y) = \text{div}\left(\overline{K_{S,+}^\times} \cap \mathbb{I}_T\right).$$

If $\mathcal{O}_{K,+}^\times$ is finite — in particular if $K = \mathbb{Q}$ or K is imaginary quadratic — then for every proper $T \subsetneq S$ the right-hand side is trivial, so $\text{Iso}_{\mathcal{G}_S}(y) = \{1\}$: the action of I_S on $\tilde{Y}_S \setminus \partial Y_S$ is free.

At the bottom stratum the isotropy is nontrivial:

$$\partial Y_S := \bigcap_{\mathfrak{a} \in J_S} \mathfrak{a} Y_S = \{[0, g]\} \cong G_S / r(U_S) \cong \text{Cl}_S^+,$$

a set of h_S^+ points, on which I_S acts transitively through the class map with isotropy $I_S^0 := \ker(I_S \rightarrow \text{Cl}_S^+)$, of index h_S^+ .

Proof. By Lemma 2.1, $\mathfrak{a} \in I_T$ lies in the isotropy iff $\sigma_{\mathfrak{a}} \in r(U_T)$, i.e. iff some idele s with $\text{div}(s) = \mathfrak{a}$ satisfies $s \in U_T \cdot \overline{K_{S,+}^\times}$. Writing $s = wu$ with $w \in U_T$ and $u \in \overline{K_{S,+}^\times}$, the element $u = w^{-1}s$ is trivial at every $\mathfrak{p}' \notin T$ and has $\text{div}(u) = \mathfrak{a}$, which is (2.2); the converse is immediate.

Suppose now $\mathcal{O}_{K,+}^\times$ is finite. Then $K_{S,+}^\times$ is discrete in $\mathbb{I}_S$: an element of $K_{S,+}^\times$ lying in the neighbourhood $\prod_{\mathfrak{p} \in F}(1 + \mathfrak{p}^k \mathcal{O}_{\mathfrak{p}}) \times \prod_{\mathfrak{p} \notin F} \mathcal{O}_{\mathfrak{p}}^\times$ of 1 has trivial divisor, hence is a totally positive unit, and is 1 for k large. So $\overline{K_{S,+}^\times} = K_{S,+}^\times$, and $u = \alpha \in K_{S,+}^\times$ is trivial at some $\mathfrak{p}' \in S \setminus T$. Under the diagonal embedding the $\mathfrak{p}'$ -component of α is α itself, and $K \rightarrow K_{\mathfrak{p}'}$ is injective, so $\alpha = 1$ and $\mathfrak{a} = (1)$.

For the bottom stratum, $\bigcap_{\mathfrak{a}} \mathfrak{a} Y_S$ forces $x_{\mathfrak{p}} = 0$ for every $\mathfrak{p}$, so $x = 0$; the fibre is $G_S / r(U_S) \cong \text{Cl}_S^+$ by (1.1), and (2.1) with $T = S$ becomes $\{\mathfrak{a} : [\mathfrak{a}] = 1\} = I_S^0$. $\square$

Remark 2.3 (The general case, and why it does not matter here). When $\mathcal{O}_{K,+}^\times$ is infinite the right-hand side of (2.2) can in principle be nontrivial: one needs α^{-1} to lie in the closure of $\mathcal{O}_{K,+}^\times$ inside $\prod_{\mathfrak{p}' \in S \setminus T} \mathcal{O}_{\mathfrak{p}'}^\times$, a condition of Leopoldt type. We do not decide it. Nothing below depends on it: Theorem 2.5 needs only that the strata are nowhere dense, which holds whatever their isotropy, since for infinite S every basic clopen set contains points with $x_{\mathfrak{p}} \neq 0$ at any prescribed $\mathfrak{p}$.

Remark 2.4. The natural expectation is that killing the coordinates in T frees the corresponding translations, giving isotropy I_T . It frees them in the $\widehat{\mathcal{O}}_S$ -direction but not in the G_S -direction: multiplication by $\mathfrak{a}$ translates g by $\sigma_{\mathfrak{a}}$, whose components at the primes *outside* T are nontrivial. Over $\mathbb{Q}$ with $T = \{2\}$ and $\mathfrak{a} = (2)$, the components of $\sigma_{\mathfrak{a}}$ at 3, 5, 7 are 2, 2, 2, none equal to 1.

Theorem 2.5 (Topological principality, and the two halves of the ideal lattice). *Let $\partial I := C^*(\mathcal{G}_S|_{\tilde{Y}_S \setminus \partial Y_S}) \cap \mathcal{A}_{K,S}$ be the ideal of $\mathcal{A}_{K,S}$ corresponding to the open invariant set $\tilde{Y}_S \setminus \partial Y_S$, so that $\mathcal{A}_{K,S} / \partial I = \partial \mathcal{A}_{K,S}$ is the boundary quotient of Theorem 3.1.*

- (1) $\mathcal{G}_S$ is topologically principal, for every K and S ; consequently $C(Y_S)$ is a Cartan subalgebra of $\mathcal{A}_{K,S}$ [18].
- (2) *The ideals of $\mathcal{A}_{K,S}$ containing ∂I correspond bijectively to the closed subsets of the torus $\widehat{I}_S^0$, via $\partial \mathcal{A}_{K,S} \cong M_{h_S^+}(C(\widehat{I}_S^0))$. These ideals are not detected by the diagonal $C(Y_S)$: two of them have the same intersection $C_0(Y_S \setminus \partial Y_S)$ with it. They are not all gauge-invariant either: the gauge action is the translation flow of $\widehat{I}_S^0$ by the characters $\mathfrak{a} \mapsto N\mathfrak{a}^{it}$, which is nontrivial, so a closed subset that is not a union of flow orbits — a point, for instance — gives an ideal that the gauge action moves.*

- (3) If the action of I_S on $\tilde{Y}_S \setminus \partial Y_S$ is free — by Proposition 2.2, whenever $\mathcal{O}_{K,+}^\times$ is finite — then the ideals of ∂I correspond bijectively and order-isomorphically to the open I_S -invariant subsets of $\tilde{Y}_S \setminus \partial Y_S$, and the diagonal detects them.
- (4) Consequently, in that case, $\text{Prim}(\mathcal{A}_{K,S}) = (\tilde{Y}_S \setminus \partial Y_S)/I_S \sqcup \hat{I}_S^0$ as a set, the first piece open and the second closed.

Proof. (1) By (2.1) every unit with nontrivial isotropy lies in some stratum $\{x_{\mathbf{p}} = 0 \mid \mathbf{p} \in T\}$ with $T \neq \emptyset$, hence in some $\{x_{\mathbf{p}} = 0\}$, a closed set with empty interior since $\mathcal{O}_{\mathbf{p}}$ has no isolated points. A countable union of such sets is meagre, so the units with trivial isotropy are dense. (This does not use Proposition 2.2.) The restriction $\mathcal{G}_S|_{Y_S}$ is then topologically principal as well, and $\mathcal{A}_{K,S} = C^*(\mathcal{G}_S|_{Y_S})$ with diagonal $C(Y_S)$, which is Cartan by Renault's theorem [18].

(2) Ideals containing ∂I are ideals of the quotient, which is $M_h(C(X))$ with $X = \hat{I}_S^0$ by Theorem 3.1; its ideals are the $M_h(C_0(X \setminus Z))$, $Z \subseteq X$ closed. The image of $C(Y_S)$ in $M_h(C(X))$ is the algebra $C(\partial Y_S) = \mathbb{C}^h$ of diagonal matrices with constant entries, which meets $M_h(C_0(X \setminus Z))$ trivially for every nonempty Z ; so every ideal $J \supseteq \partial I$ with $J \neq \mathcal{A}_{K,S}$ has $J \cap C(Y_S) = \partial I \cap C(Y_S) = C_0(Y_S \setminus \partial Y_S)$. The gauge action $\sigma_t(\mu_{\mathbf{a}}) = \text{Na}^{it}\mu_{\mathbf{a}}$ descends to the unitaries $\mu_{\mathbf{a}}$, $\mathbf{a} \in I_S^0$, of $C^*(I_S^0) = C(X)$, where it is translation by the character $\chi_t : \mathbf{a} \mapsto \text{Na}^{it}$; $\chi_t \neq 1$ for $t \neq 0$ since $\log \text{Na} \neq 0$ for $\mathbf{a} \neq (1)$, so the flow moves every point of X .

(3) $\mathcal{G}_S|_{\tilde{Y}_S \setminus \partial Y_S}$ is second countable, étale, amenable and principal, so by Renault [17] its C^* -algebra has ideals in bijection with the open invariant subsets, via $V \mapsto C^*(\mathcal{G}_S|_V)$, and its diagonal is a Cartan subalgebra detecting them; ∂I is the full corner $\mathbf{1}_{Y_S}(\cdot)\mathbf{1}_{Y_S}$, which preserves the lattice. (4) is (2) and (3) together with the exactness of $0 \rightarrow C^*(\mathcal{G}_S|_U) \rightarrow C^*(\mathcal{G}_S) \rightarrow C^*(\mathcal{G}_S|_{U^c}) \rightarrow 0$ for amenable étale groupoids. $\square$

Remark 2.6. Part (2) shows that “the ideal lattice is the lattice of open invariant sets” is false for $\mathcal{A}_{K,S}$: the boundary orbit has isotropy I_S^0 of infinite rank, and its C^* -algebra $C(\hat{I}_S^0)$ contributes a torus of primitive ideals invisible to the diagonal. Topological principality of $\mathcal{G}_S$ gives the correspondence only on the essentially principal part $\tilde{Y}_S \setminus \partial Y_S$. When $\mathcal{O}_{K,+}^\times$ is infinite, Remark 2.3 leaves open whether some proper stratum has nontrivial isotropy; if it does, that stratum contributes further ideals of the same kind.

Proposition 2.7 (The lattice is not the power set of S). *Let $K = \mathbb{Q}$, $S = \{\ell\}$, so $Y_S = \mathbb{Z}_\ell$ and $\mathcal{A} = C(\mathbb{Z}_\ell) \rtimes \mathbb{N}$. Then $\mathcal{A}$ is a full corner in $C_0(\mathbb{Q}_\ell) \rtimes \mathbb{Z}$ and there is an exact sequence*

$$(2.3) \quad 0 \longrightarrow C(\mathbb{Z}_\ell^\times) \otimes \mathcal{K} \longrightarrow C_0(\mathbb{Q}_\ell) \rtimes \mathbb{Z} \longrightarrow C(\mathbb{T}) \longrightarrow 0.$$

The ideal lattice of $\mathcal{A}$ contains an order-isomorphic copy of the lattice of open subsets of the Cantor set $\mathbb{Z}_\ell^\times$, hence is uncountable.

Proof. By Laca's dilation theorem $\mathcal{A}$ is a full corner in $C_0(\mathbb{Q}_\ell) \rtimes \mathbb{Z}$, the dilation being the limit of $\mathbb{Z}_\ell \xleftarrow{\ell} \mathbb{Z}_\ell \xleftarrow{\ell} \cdots$. On $\mathbb{Q}_\ell \setminus \{0\}$ the action of $\mathbb{Z}$ by multiplication by ℓ is free and proper with quotient $\mathbb{Z}_\ell^\times$, giving the ideal in (2.3); the fixed point $\{0\}$ gives the quotient $C^*(\mathbb{Z}) = C(\mathbb{T})$. Ideals of $C(\mathbb{Z}_\ell^\times) \otimes \mathcal{K}$ correspond to open subsets of $\mathbb{Z}_\ell^\times$ and pass to $\mathcal{A}$ by the corner correspondence. $\square$

Remark 2.8. The description “hereditary subsets of $\mathcal{P}(S)$ ” would be correct for a graph or Cuntz–Krieger algebra. Here the stratification by vanishing coordinates is present, but each stratum retains a Cantor set of unit orbits, each closed set of orbits contributes an ideal, and the quotient $C(\mathbb{T})$ in (2.3) is the $h_S^+ = 1$, $I_S^0 = \mathbb{Z}$ case of Theorem 2.5(2).

3. THE BOUNDARY QUOTIENT

Theorem 3.1 (The tight quotient is abelian-by-finite). *Let $\partial\mathcal{A}_{K,S}$ be the quotient by the ideal corresponding to $\tilde{Y}_S \setminus \partial Y_S$, equivalently the tight quotient in the sense of Exel [5], equivalently the*

universal quotient in which every μ_p becomes unitary. Then

$$\partial\mathcal{A}_{K,S} \cong C(\text{Cl}_S^+) \rtimes I_S \cong M_{h_S^+}(C^*(I_S^0)) \cong M_{h_S^+}(C(\widehat{I_S^0})).$$

It is unital, separable, nuclear, satisfies the UCT, is of type I and carries a faithful tracial state. It is not simple when $I_S^0 \neq 0$, and is never purely infinite; in particular it is not a Kirchberg algebra.

Proof. Restricting the groupoid to the closed invariant set ∂Y_S gives $C^*(I_S \ltimes \partial Y_S) = C(\text{Cl}_S^+) \rtimes I_S$ by Proposition 2.2. In the quotient $\mathbf{1}_{pY_S}$ restricts to 1 on $\partial Y_S \subseteq pY_S$, so μ_p is unitary; conversely imposing unitarity kills exactly the functions vanishing on ∂Y_S . For a discrete group Γ with a subgroup Γ_0 of finite index h , Green's imprimitivity theorem [7] gives $C(\Gamma/\Gamma_0) \rtimes \Gamma \cong M_h(C^*(\Gamma_0))$, and I_S^0 is free abelian so $C^*(I_S^0)$ is commutative. The remaining properties are those of $M_h(C(X))$; a unital algebra with a tracial state is never purely infinite. $\square$

Remark 3.2 (Why no Cuntz relation is available). By Lemma 1.1(2) the projections $\mathbf{1}_{aY_S}$ are nested. Imposing $\sum_a \mu_a \mu_a^* = 1$ over any finite family would force all but one summand to vanish. The tight relation therefore degenerates to $\mathbf{1}_{pY_S} = 1$, which is unitarity, and this is what Theorem 3.1 implements. The Cuntz alternative requires a partition; see §6.

Theorem 3.3 (The class of the unit records h_S^+). $K_0(\partial\mathcal{A}_{K,S}) \cong \Lambda^{\text{ev}}(I_S^0)$ and $K_1(\partial\mathcal{A}_{K,S}) \cong \Lambda^{\text{odd}}(I_S^0)$, and under this identification

$$(3.1) \quad [1_{\partial\mathcal{A}_{K,S}}] = h_S^+ \cdot e_0,$$

e_0 the generator of $\Lambda^0 = \mathbb{Z}$. Hence h_S^+ is the largest integer dividing $[1]$, and is an isomorphism invariant of the pair $(\partial\mathcal{A}_{K,S}, [1])$.

Proof. K -theory is Morita invariant and $C^*(\Gamma) \cong C(\widehat{\Gamma})$ for Γ free abelian has $K_0 = \Lambda^{\text{ev}}(\Gamma)$, $K_1 = \Lambda^{\text{odd}}(\Gamma)$ with $[1]$ the generator of Λ^0 . The isomorphism $K_0(M_h(B)) \cong K_0(B)$ sends $[1_{M_h(B)}]$ to $h[1_B]$, since $1_{M_h(B)}$ is a sum of h orthogonal projections each equivalent to 1_B ; and Λ^{ev} is free abelian, so $h e_0$ is divisible by exactly h . $\square$

Theorem 3.4 (Rigidity of the boundary). For infinite $S_1 \subseteq P_{K_1}$ and $S_2 \subseteq P_{K_2}$,

$$\partial\mathcal{A}_{K_1,S_1} \cong \partial\mathcal{A}_{K_2,S_2} \iff h_{S_1}^+ = h_{S_2}^+.$$

The boundary determines the order of the subgroup of the narrow class group generated by S , and nothing else: not K , not S , not Cl_S^+ as a group.

Proof. ($\Leftarrow$) For infinite S , I_S is free abelian of countably infinite rank and I_S^0 has finite index, hence the same rank; so the two algebras are $M_h(C(X))$ with the same h and homeomorphic X . ($\Rightarrow$) Every irreducible representation of $M_h(C(X))$ has dimension exactly h , and isomorphisms preserve irreducible representations and their dimensions; alternatively apply Theorem 3.3. For the last statement, $C(\text{Cl}_S^+) \rtimes I_S$ depends on Cl_S^+ only through the index of I_S^0 . $\square$

4. K -THEORY OF $\mathcal{A}_{K,S}$

The Stacey crossed product imposes the covariance relation $\mu_p \mu_p^* = \alpha_p(1)$, which is Katsura covariance; hence for a single prime $\mathcal{A}_{K,S}$ is the Cuntz–Pimsner algebra $\mathcal{O}_X$ of the associated correspondence, not the Toeplitz algebra $\mathcal{T}_X$. Pimsner's KK -equivalence $\iota : A \rightarrow \mathcal{T}_X$ therefore does not apply, and one must use his exact sequence.

Lemma 4.1. Y_S is compact, metrizable and totally disconnected without isolated points, hence a Cantor set. Consequently $K^0(Y_S) = C(Y_S, \mathbb{Z})$ is free abelian of countably infinite rank and $K^1(Y_S) = 0$.

Proof. $\widehat{\mathcal{O}}_S$ is profinite, G_S is a quotient of the totally disconnected $\mathbb{I}_S$ by a closed subgroup hence profinite, and Y_S is their quotient by the compact group U_S . A point $[x, g]$ is not isolated: every neighbourhood contains $[x', g]$ with $x' \neq x$, since no $\mathcal{O}_{\mathfrak{p}}$ has isolated points. $\square$

Proposition 4.2 (One prime). *Let $S = \{\mathfrak{p}\}$ and let α_* be the endomorphism $\mathbf{1}_U \mapsto \mathbf{1}_{\mathfrak{p}U}$ of $K^0(Y_S) = C(Y_S, \mathbb{Z})$. Then*

$$(4.1) \quad K_1(\mathcal{A}_{K,S}) = \ker(1 - \alpha_*) = 0, \quad K_0(\mathcal{A}_{K,S}) = \operatorname{coker}(1 - \alpha_*) \cong C(G_S, \mathbb{Z})/\mathbb{Z} \cdot 1 \oplus \mathbb{Z},$$

a free abelian group. In particular $[1] = [\mathbf{1}_{Y_S}]$ in K_0 , so $\iota_ : K^0(Y_S) \rightarrow K_0(\mathcal{A}_{K,S})$ is not injective.*

Proof. $\mu_{\mathfrak{p}}$ is an isometry with $\mu_{\mathfrak{p}}\mu_{\mathfrak{p}}^* = \mathbf{1}_{Y_S}$, so $1 \sim \mathbf{1}_{Y_S}$ and $[1] = [\mathbf{1}_{Y_S}]$. The six-term sequence for the Cuntz–Pimsner algebra of a correspondence [14, §4] (see [10] for the covariance condition) with $K_1(C(Y_S)) = 0$ collapses to $K_1(\mathcal{A}) = \ker(1 - \alpha_*)$, $K_0(\mathcal{A}) = \operatorname{coker}(1 - \alpha_*)$ on $C(Y_S, \mathbb{Z})$.

If $f = \alpha_* f = (f \circ \mathfrak{p}^{-1})\mathbf{1}_{Y_S}$ then f is supported in $\mathfrak{p}^n Y_S$ for every n , i.e. in $\partial Y_S = \{[0, g]\}$, a closed set with empty interior; a locally constant function supported there vanishes. So $K_1 = 0$.

For K_0 we use Lacas's dilation [12]: $\mathcal{A}$ is a full corner in $C_0(\tilde{Y}_S) \rtimes \mathbb{Z}$, and by Pimsner–Voiculescu [15] $K_0(C_0(\tilde{Y}_S) \rtimes \mathbb{Z}) = H_0(\mathbb{Z}; C_c(\tilde{Y}_S, \mathbb{Z}))$, the coinvariants, $K_1 = H_1$, the invariants. Write $\tilde{Y}_S = \tilde{Y}_\emptyset \sqcup \partial Y_S$ with $\tilde{Y}_\emptyset = \{[x, g] : x \neq 0\}$ open. The map $[\pi^n u, g] \mapsto (n, r(u)g)$ identifies $\tilde{Y}_\emptyset$ with $\mathbb{Z} \times G_S$, on which $\mathfrak{p}$ acts by $(n, h) \mapsto (n+1, \sigma_{\mathfrak{p}} h)$, freely; hence $H_1(\mathbb{Z}; C_c(\tilde{Y}_\emptyset, \mathbb{Z})) = 0$ and $H_0 = C(G_S, \mathbb{Z})$, the class of f being $\sum_n (\sigma_{\mathfrak{p}}^{-n})_* f(n, \cdot)$. On $\partial Y_S \cong \operatorname{Cl}_S^+$ the generator acts through the class map, so $H_1 = H_0 = \mathbb{Z}$ by Shapiro's lemma. The long exact sequence of $0 \rightarrow C_c(\tilde{Y}_\emptyset, \mathbb{Z}) \rightarrow C_c(\tilde{Y}_S, \mathbb{Z}) \rightarrow C(\partial Y_S, \mathbb{Z}) \rightarrow 0$ reads

$$0 \rightarrow H_1(\tilde{Y}_S) \rightarrow \mathbb{Z} \xrightarrow{\delta} C(G_S, \mathbb{Z}) \rightarrow H_0(\tilde{Y}_S) \rightarrow \mathbb{Z} \rightarrow 0,$$

and $\delta(1)$ is the class of $(1 - \alpha_*)\mathbf{1}_{Y_S} = \mathbf{1}_{Y_S \setminus \mathfrak{p}Y_S} = \mathbf{1}_{\{0\} \times G_S}$, i.e. the constant function 1; so δ is injective, $H_1(\tilde{Y}_S) = 0$ again, and $H_0(\tilde{Y}_S)$ is an extension of $\mathbb{Z}$ by $C(G_S, \mathbb{Z})/\mathbb{Z} \cdot 1$, which splits. Both summands are free, 1 being part of a basis of $C(G_S, \mathbb{Z})$. $\square$

For S finite with $|S| \geq 2$ the primes interact: by [21, Lem. 2.10] the generator $\mathfrak{p}$ moves the G_S -coordinate by $\sigma_{\mathfrak{p}}$, whose components at the other primes are nontrivial, so the action of $\mathbb{Z}^S$ on $\tilde{Y}_S$ is not a product of actions on the factors $K_{\mathfrak{p}}$. What survives is the following.

Proposition 4.3 (Finite S). *For S finite, $K_*(\mathcal{A}_{K,S}) = K_*(C_0(\tilde{Y}_S) \rtimes \mathbb{Z}^S)$ is computed by the spectral sequence $E_{p,q}^2 = H_p(\mathbb{Z}^S; K_q(C_0(\tilde{Y}_S))) \Rightarrow K_{p+q}$ of Kasparov [9, §6]. Since $K_1(C_0(\tilde{Y}_S)) = 0$, only the row $q = 0$ is nonzero, the sequence collapses, and $K_0(\mathcal{A}_{K,S})$ (resp. K_1) carries a finite filtration whose subquotients are the even (resp. odd) Koszul homology groups $H_p(\mathbb{Z}^S; C_c(\tilde{Y}_S, \mathbb{Z}))$ of the commuting automorphisms $\alpha_{\mathfrak{p}*}$, $\mathfrak{p} \in S$. The bottom piece $H_0(\mathbb{Z}^S; C_c(\tilde{Y}_S, \mathbb{Z}))$, the coinvariants, is the image of $K^0(Y_S) \rightarrow K_0(\mathcal{A}_{K,S})$. For general S , $K_*(\mathcal{A}_{K,S}) = \varinjlim K_*(\mathcal{A}_{K,F})$ over finite $F \subseteq S$, for any compatible system of embeddings as in [21, Prop. 2.7] (see Remark 5.1 below).*

Proof. Lacas's dilation, Kasparov's spectral sequence for actions of $\mathbb{Z}^n$, and the continuity of K -theory under direct limits. $\square$

We have not computed the Koszul homology for $|S| \geq 2$, and we make no claim about $K_*(\mathcal{A}_{K,S})$ beyond Propositions 4.2 and 4.3; the top group $H_{|S|}$ is the group of $\mathbb{Z}^S$ -invariant compactly supported functions, which vanishes by the argument of Proposition 4.2, but the intermediate groups receive contributions from the strata $\{x_{\mathfrak{p}} = 0, \mathfrak{p} \in T\}$, on which $\mathbb{Z}^T$ acts on G_S by translations with nontrivial invariants.

Example 4.4 ($K = \mathbb{Q}$, $S = \{\ell\}$). Here $G_S = \mathbb{Z}_{\ell}^{\times}$ and Proposition 4.2 gives $K_1(\mathcal{A}) = 0$ and $K_0(\mathcal{A}) \cong C(\mathbb{Z}_{\ell}^{\times}, \mathbb{Z})/\mathbb{Z} \oplus \mathbb{Z} \cong \mathbb{Z}^{(\infty)}$, in agreement with (2.3). A direct computation of the Smith

normal form of $1 - \alpha_*$ on level- n functions, for $\ell = 2, 3, 5$ and $n \leq 3$, gives kernel 0 and free cokernel of rank $\ell^{n+1} - \ell^n$, with no torsion.

Remark 4.5 (K -theory carries no arithmetic: the mechanism). For a single prime, $K_0(\mathcal{A}_{K,S})$ is free abelian and depends on $(K, \mathfrak{p})$ only through the profinite group G_S ; for $K = \mathbb{Q}$ it is $\mathbb{Z}^{(\infty)}$ for every ℓ . The reason is structural rather than accidental: by Propositions 4.2 and 4.3 the image of $K^0(Y_S)$ in $K_0(\mathcal{A}_{K,S})$ is the group of coinvariants $C(Y_S, \mathbb{Z}) / \sum_{\mathfrak{p}} (1 - \alpha_{\mathfrak{p}*})C(Y_S, \mathbb{Z})$, and a functional on $C(Y_S, \mathbb{Z})$ descends to it exactly when it is α_* -invariant, whereas the arithmetically meaningful functionals — the KMS_β measures — satisfy $\omega \circ \alpha_{\mathfrak{p}*} = N\mathfrak{p}^{-\beta}\omega$ with $N\mathfrak{p}^{-\beta} \neq 1$. The passage to K_0 annihilates precisely the data of interest. This is taken up in Paper III of the series.

5. INDEX THEORY

Remark 5.1 (Nested prime sets). For $S_1 \subset S_2$ the embedding $\mathcal{A}_{K,S_1} \hookrightarrow \mathcal{A}_{K,S_2}$, $f \mapsto f \circ \text{pr}$, $\mu_{\mathfrak{a}} \mapsto \mu_{\mathfrak{a}}$, of [21, Prop. 2.7] requires an I_{S_1} -equivariant continuous surjection $\text{pr} : Y_{S_2} \rightarrow Y_{S_1}$. By (1.1), Y_S is a disjoint union over Cl_S^+ of copies of $\widehat{\mathcal{O}}_S / \overline{\mathcal{O}_{K,+}^\times}$, and $\text{Cl}_{S_1}^+$ is a *subgroup* of $\text{Cl}_{S_2}^+$, not a quotient. The coordinate projection $\widehat{\mathcal{O}}_{S_2} \rightarrow \widehat{\mathcal{O}}_{S_1}$ therefore descends canonically exactly when $\text{Cl}_{S_2}^+ = \text{Cl}_{S_1}^+$, i.e. when the primes of $S_2 \setminus S_1$ have classes in $\text{Cl}_{S_1}^+$; this holds for $K = \mathbb{Q}$, for every S_1 generating the narrow class group, and for all $S_1 \subseteq S_2$ inside a fixed S once S_1 is large enough. Otherwise pr exists but involves the choice of a $\text{Cl}_{S_1}^+$ -equivariant retraction $\text{Cl}_{S_2}^+ \rightarrow \text{Cl}_{S_1}^+$, and the words “ $G_S = \varprojlim G_{S_k}$ ” in [21, Prop. 2.7] must be read with this qualification (the uses made of that proposition in [21] are over $\mathbb{Q}$). *Throughout this section we assume $\text{Cl}_{S_2}^+ = \text{Cl}_{S_1}^+$, so that pr is canonical.*

Let $S_1 \subset S_2$, let φ_2 be a KMS_β state of $\mathcal{A}_{K,S_2}$ with $\zeta_{K,S_2}(\beta) = \infty$, and put $M_{S_i} = \pi_{\varphi_2}(\mathcal{A}_{K,S_i})''$, where $\mathcal{A}_{K,S_1} \subseteq \mathcal{A}_{K,S_2}$ via Remark 5.1.

Proposition 5.2 (The conditional expectation exists). $\sigma_t^{\varphi_2} = \sigma_{-\beta t}$ leaves $\mathcal{A}_{K,S_1}$ globally invariant, so by Takesaki’s theorem [19] there is a unique faithful normal conditional expectation $E : M_{S_2} \rightarrow M_{S_1}$ with $\varphi_2 \circ E = \varphi_2$. If φ_2 is extremal, its restriction to $\mathcal{A}_{K,S_1}$ need not be, so M_{S_1} need not be a factor.

Proof. σ_t fixes $C(Y_{S_2}) \supseteq C(Y_{S_1})$ pointwise and scales each $\mu_{\mathfrak{a}}$, so preserves $\mathcal{A}_{K,S_1}$; the GNS vector of a KMS state is separating, so φ_2 is faithful on M_{S_2} with modular group $\sigma_{-\beta t}$. Extremality is not preserved by restriction because the simplices are indexed by $\widehat{\Xi_\beta(S_2)}$ and $\widehat{\Xi_\beta(S_1)}$ respectively. $\square$

We write $\text{Ind}(F) \in [1, \infty]$ for the Pimsner–Popa index of a faithful normal conditional expectation $F : M_{S_2} \rightarrow M_{S_1}$, the inverse of the best constant λ with $F(x) \geq \lambda x$ for all $x \geq 0$ [16], and $[M_{S_2} : M_{S_1}]_0 = \inf_F \text{Ind}(F)$ for the Kosaki–Longo index [11, 13].

Theorem 5.3 (Prime inclusions have infinite index). Let $\mathfrak{p} \in S_2 \setminus S_1$ and $e_k = \mathbf{1}_{\{v_{\mathfrak{p}}=k\}} = \mathbf{1}_{\mathfrak{p}^k Y_{S_2}} - \mathbf{1}_{\mathfrak{p}^{k+1} Y_{S_2}}$.

- (1) For every KMS_β state φ_2 , the e_k are pairwise orthogonal projections in $M'_{S_1} \cap M_{S_2}$ with $\sum_k e_k = 1$ and $\varphi_2(e_k) = (1 - N\mathfrak{p}^{-\beta})N\mathfrak{p}^{-k\beta} > 0$; in particular $M'_{S_1} \cap M_{S_2}$ contains a copy of $\ell^\infty(\mathbb{N})$.
- (2) Let $\varphi_2 = \varphi_{\text{sym}}$ be the symmetric KMS_β state, so that in the coordinates of [21, Lem. 2.10] its measure is $\nu = \pi_\beta \otimes \text{Haar}$. Then $\text{Ind}(F) = \infty$ for every faithful normal conditional expectation $F : M_{S_2} \rightarrow M_{S_1}$, hence $[M_{S_2} : M_{S_1}]_0 = \infty$, already when a single prime is added.

Proof. (1) For $\mathfrak{q} \in S_1$, multiplication by $\mathfrak{q}$ does not change $v_{\mathfrak{p}}$, so $\mathfrak{q}\{v_{\mathfrak{p}} \geq k\} = \{v_{\mathfrak{p}} \geq k\} \cap \mathfrak{q}Y_{S_2}$ and

$$\mu_{\mathfrak{q}} \mathbf{1}_{\{v_{\mathfrak{p}} \geq k\}} = \alpha_{\mathfrak{q}}(\mathbf{1}_{\{v_{\mathfrak{p}} \geq k\}}) \mu_{\mathfrak{q}} = \mathbf{1}_{\{v_{\mathfrak{p}} \geq k\}} \mathbf{1}_{\mathfrak{q}Y_{S_2}} \mu_{\mathfrak{q}} = \mathbf{1}_{\{v_{\mathfrak{p}} \geq k\}} \mu_{\mathfrak{q}},$$

using $\mathbf{1}_{qY}\mu_q = \mu_q\mu_q^*\mu_q = \mu_q$; taking adjoints gives commutation with μ_q^* , and commutation with $C(Y_{S_1})$ is automatic. The $\mathbf{1}_{\{v_p \geq k\}}$ therefore commute with $\mathcal{A}_{K,S_1}$, hence with M_{S_1} , and so do their differences e_k . The values $\varphi_2(e_k)$ come from $\varphi_2(\mathbf{1}_{p^kY}) = Np^{-k\beta}$ [21, Prop. 2.8], and $\sum_k e_k = \mathbf{1}_{\{v_p < \infty\}} = 1$ in M_{S_2} because $\varphi_2(v_p \geq k) \rightarrow 0$.

(2) Let $\varphi_2 = \varphi_{\text{sym}}$. The GNS space is $H = L^2(\mathcal{R}, \tilde{\nu})$, $\mathcal{R}$ the orbit relation of I_{S_2} on (Y_{S_2}, ν) and $\tilde{\nu}$ its lift, $M_{S_2} = W^*(\mathcal{R}, \nu)$, and e_k acts by multiplication by $\mathbf{1}_{\{v_p = k\}}$ in the first variable [21, Prop. 6.1]. Fix $k, j \geq 0$ and let $T : \{v_p = k\} \rightarrow \{v_p = j\}$, $T(v, h) = (v + (j - k)e_p, h)$, in the coordinates $(v, h) \in V \times G_{S_2}$: T moves the p -valuation and nothing else. It is a homeomorphism between clopen sets with $\nu \circ T = Np^{-(j-k)\beta}\nu$ on its domain, since ν is a product measure with geometric p -marginal and Haar G_{S_2} -marginal; and it commutes with the maps $(v, h) \mapsto (v + e_q, \sigma_q h)$, $q \in S_1$, generating the action of I_{S_1} . Hence T maps $\mathcal{R}$ -classes to $\mathcal{R}$ -classes, and

$$(W\xi)(y, y') = Np^{(j-k)\beta/2} \xi(T^{-1}y, T^{-1}y'), \quad y \in \{v_p = j\},$$

is a partial isometry on H with $W^*W = e_k$ and $WW^* = e_j$. Because T commutes with the I_{S_1} -action, does not move the coordinates seen by $C(Y_{S_1}) \circ \text{pr}$, and has constant Radon–Nikodym derivative, W commutes with $\pi(\mathcal{A}_{K,S_1})$, hence $W \in M'_{S_1}$ (the commutant in $B(H)$; W need not lie in M_{S_2}).

Now let F be a faithful normal conditional expectation with $\text{Ind}(F) = \lambda^{-1} < \infty$, so $F(e_k) \geq \lambda e_k$. Since e_k commutes with M_{S_1} , $F(e_k)$ lies in $Z(M_{S_1})$ and commutes with e_k ; on the range of e_k it is $\geq \lambda$, so $e_k \leq q_k := \mathbf{1}_{[\lambda, 1]}(F(e_k)) \in M_{S_1}$, and $\lambda q_k \leq F(e_k)$ gives $\varphi_2(q_k) \leq \lambda^{-1}\varphi_2(F(e_k))$. For $q \in M_{S_1}$ with $qe_k = e_k$ we get, using $W \in M'_{S_1}$,

$$qe_j = qWW^* = WqW^* = W(qe_k)W^* = We_kW^* = e_j \quad \text{for every } j,$$

so $q = \sum_j e_j q = 1$. Applied to $q = q_k$: $1 = \varphi_2(q_k) \leq \lambda^{-1}\varphi_2(F(e_k))$, i.e. $\varphi_2(F(e_k)) \geq \lambda$ for every k . But $F(e_k) \in Z(M_{S_1})$ and $\sum_k F(e_k) = 1$, so $\sum_k \varphi_2(F(e_k)) = 1$, and $\varphi_2(F(e_k)) \geq \lambda > 0$ for infinitely many k is impossible. $\square$

Remark 5.4 (Scope). Part (1) holds for every KMS state; part (2) uses the product form of ν only to build W . For a non-symmetric extremal state the fibre measure ρ_v of [21, Lem. 2.10] is Haar measure on a coset that moves by σ_p when v_p increases, so T no longer preserves the measure class, and we have not decided whether every expectation has infinite index in that case; the symmetric state is extremal, and M_{S_2} a factor, precisely when $\Xi_\beta(S_2) = \{1\}$, e.g. for $S_2 = P_K$ and $\beta \leq 1$.

Remark 5.5 (Why no index formula can exist). An index formula in Np , a local unit index and $[Cl_{S_2}^+ : Cl_{S_1}^+]$ would combine finite quantities. It cannot exist because enlarging S enlarges the *base space*: Y_{S_2} carries a new p -adic coordinate, and the valuation v_p alone, before any unit or class-group data enters, already contributes an infinite-dimensional commuting algebra. Those finite invariants do appear in this theory — in Theorem 3.3 and in (1.1) — but not in an index.

Proposition 5.6 (No Galois correspondence with prime sets). *In the situation of Theorem 5.3(2) let $S_2 = S_1 \cup \{p\}$, so there are exactly two intermediate prime sets. Then $\text{Lat}(M_{S_1} \subset M_{S_2})$ has cardinality at least 2^{N_0} .*

Proof. For a partition $\mathcal{P}$ of $\mathbb{N}$ let $A_{\mathcal{P}} \subseteq W^*(e_k : k \geq 0) \cong \ell^\infty(\mathbb{N})$ be the subalgebra of functions constant on the blocks; it lies in the relative commutant, so $N_{\mathcal{P}} = M_{S_1} \vee A_{\mathcal{P}}$ is intermediate. We recover $\mathcal{P}$ from $N_{\mathcal{P}}$: an element $x = \sum_k c_k e_k$ of $W^*(e_k)$ lies in $N_{\mathcal{P}}$ iff for every block B the element $\sum_{k \in B} c_k e_k$ lies in $M_{S_1} \mathbf{1}_B$, i.e. equals $m \mathbf{1}_B$ for some $m \in M_{S_1}$; conjugating by the partial isometries W of the proof of Theorem 5.3 gives $c_j e_j = m e_j = W(m e_k) W^* = c_k e_j$ for $j, k \in B$, so $W^*(e_k) \cap N_{\mathcal{P}} = A_{\mathcal{P}}$. There are 2^{N_0} partitions. $\square$

We now locate the finite index. Fix S and β with Ξ_β finite, and write $G := \widehat{\Xi_\beta}$, φ_{sym} for the G_S -invariant KMS_β state, and φ_ε , $\varepsilon \in G$, for the extreme points.

Lemma 5.7 (Central decomposition).

$$(5.1) \quad M_{\text{sym}} := \pi_{\varphi_{\text{sym}}}(\mathcal{A}_{K,S})'' \cong \bigoplus_{\varepsilon \in G} M_\varepsilon, \quad M_\varepsilon := \pi_{\varphi_\varepsilon}(\mathcal{A}_{K,S})'',$$

with centre $Z(M_{\text{sym}}) \cong \mathbb{C}^G$ spanned by projections z_ε , each M_ε a factor, and $\omega_{\varphi_{\text{sym}}}(z_\varepsilon) = |G|^{-1}$. The action θ of G_S extends to M_{sym} , factors through $G_S/\Xi_\beta^\perp \cong G$, and permutes the z_ε simply transitively.

Proof. By [21, Thm. 3.6] the KMS_β simplex is $\text{Prob}(G_S/\Xi_\beta^\perp)$, so its extreme points are indexed by $G = \widehat{\Xi_\beta}$ and correspond to mutually singular measures ν_ε ; hence the GNS representations are pairwise disjoint and each M_ε is a factor. The symmetric state is the uniform mixture $\varphi_{\text{sym}} = |G|^{-1} \sum_\varepsilon \varphi_\varepsilon$, so $\pi_{\varphi_{\text{sym}}} = \bigoplus_\varepsilon \pi_{\varphi_\varepsilon}$, giving (5.1), the centre, and $\omega_{\varphi_{\text{sym}}}(z_\varepsilon) = |G|^{-1}$. By [21, Thm. 3.6(3)] the action of G_S on the simplex factors through $G_S/\Xi_\beta^\perp$ and permutes the extreme points simply transitively; it therefore permutes the z_ε the same way. $\square$

Theorem 5.8 (The finite-index inclusion). Put $D := M_{\varepsilon_0}$ for any $\varepsilon_0 \in G$. The isomorphisms $\theta_g : M_{\varepsilon_0} \rightarrow M_{g\varepsilon_0}$ identify the inclusion $M_{\text{sym}}^G \subseteq M_{\text{sym}}$ with

$$(5.2) \quad D \otimes 1 \subseteq D \otimes \ell^\infty(G) = \bigoplus_{g \in G} D.$$

Consequently $M_{\text{sym}}^G \cong D$ is a factor,

$$(5.3) \quad [M_{\text{sym}} : M_{\text{sym}}^G]_0 = |G| = |\Xi_\beta|, \quad M_{\text{sym}}^G \cap M_{\text{sym}} = Z(M_{\text{sym}}) \cong \mathbb{C}^{|G|},$$

the index being realized by $E(x) = |G|^{-1} \sum_g \theta_g(x)$; and the intermediate von Neumann algebras are exactly the $D \otimes \ell^\infty(G/\mathcal{P})$, $\mathcal{P}$ a partition of G , so $\text{Lat}(M_{\text{sym}}^G \subset M_{\text{sym}})$ is the partition lattice of G . The subgroup lattice of G embeds in it, order-reversingly, by $H \mapsto M_{\text{sym}}^H$, which corresponds to the partition of G into cosets of H ; for $|G| \geq 3$ the embedding is proper.

Proof. G permutes the summands of (5.1) simply transitively, so an invariant element is determined by one component and $M_{\text{sym}}^G \cong D$ embedded diagonally via the θ_g , which is (5.2). In (5.2) the averaging map is $D \otimes E_0$ with E_0 the uniform state on $\ell^\infty(G)$, whose Pimsner–Popa index is $|G|$; any other expectation is $D \otimes E_\mu$ with μ a faithful probability on G and index $\max_g \mu(g)^{-1} \geq |G|$, so the Kosaki–Longo index is $|G|$. An element of M_{sym} commuting with the diagonal copy of the factor D is scalar in each summand, giving $\mathbb{C}^{|G|}$. By the Ge–Kadison splitting theorem [6], every von Neumann algebra between $D \otimes 1$ and $D \otimes \ell^\infty(G)$ is $D \otimes B$ for a von Neumann subalgebra $B \subseteq \ell^\infty(G)$, and the unital subalgebras of $\ell^\infty(G)$ are the $\ell^\infty(G/\mathcal{P})$. The H -invariant elements are those constant on the H -orbits, i.e. on the cosets of H ; the partition $\{1\}, G \setminus \{1\}$ of a group of order ≥ 3 is not a coset partition. $\square$

Remark 5.9 (Not a subfactor invariant). The inclusion (5.2) is the trivial inclusion of the factor D tensored with $\mathbb{C} \subseteq \ell^\infty(G)$: its Jones tower is $D \otimes (\mathbb{C} \subseteq \ell^\infty(G) \subseteq M_{|G|}(\mathbb{C}) \subseteq \dots)$, and its intermediate algebras are governed by partitions rather than by subgroups. It is therefore *not* of the form $M^G \subset M$ for an outer action on a factor, and it carries no fusion category of bimodules beyond the data of the finite set G . The arithmetic content of the symmetry breaking is entirely in the number $|\Xi_\beta|$ and in the way G_S permutes the summands; this is what Corollary 5.11 quantifies. Along a cascade with $\Xi_\beta \cong (\mathbb{Z}/2\mathbb{Z})^{A_\beta}$ [21, Prop. 4.16], crossing a threshold β_n doubles the number of summands and doubles the index.

Lemma 5.10 (Relative entropy of the phases). *With the notation of Lemma 5.7, and S the relative entropy,*

$$S(\omega_{\varphi_\varepsilon} \parallel \omega_{\varphi_{\text{sym}}}) = \log |G|,$$

and the whole value is carried by the centre: the contribution of each summand M_ε vanishes. For $\varepsilon \neq \varepsilon'$ one has $S(\omega_{\varphi_\varepsilon} \parallel \omega_{\varphi_{\varepsilon'}}) = +\infty$.

Proof. For a direct sum $M = \bigoplus_i M_i$ with normal states $\omega = \bigoplus \lambda_i \omega_i$ and $\psi = \bigoplus \mu_i \psi_i$,

$$S(\omega \parallel \psi) = \sum_i \lambda_i \log \frac{\lambda_i}{\mu_i} + \sum_i \lambda_i S(\omega_i \parallel \psi_i),$$

with the conventions $0 \log 0 = 0$ and $\lambda \log(\lambda/0) = +\infty$ for $\lambda > 0$. Apply this to (5.1) with $\lambda = \delta_\varepsilon$ and $\mu_\delta = |G|^{-1}$ for every δ , which is Lemma 5.7. On the summand ε the two states induce the same normal state of M_ε , so that term vanishes; on the others $\lambda_\delta = 0$. Hence $S = \log |G|$ and the second sum is identically zero, which is the assertion that the relative entropy is purely central. For $\varepsilon \neq \varepsilon'$ the two states are supported on orthogonal central projections and the first sum is $\log(1/0) = +\infty$. $\square$

Corollary 5.11 (Index staircase equals information staircase).

$$\log [M_{\text{sym}} : M_{\text{sym}}^G]_0 = |A_\beta| \log 2 = S(\omega_{\varphi_\varepsilon} \parallel \omega_{\varphi_{\text{sym}}}),$$

by (5.3) and Lemma 5.10. *The subfactor index and the relative entropy are the same staircase.*

The coincidence is not an accident, but neither is it deep: both sides count phases. The index of $M_{\text{sym}}^G \subset M_{\text{sym}}$ is the number $|G|$ of extremal KMS_β states, by (5.2); and the relative entropy of one extremal state against the uniform mixture of $|G|$ mutually singular states is $\log |G|$, the information needed to name which phase one is in. Each threshold β_n doubles the number of phases, so each adds $\log 2$ to both quantities. What the identity does say is that the operator-algebraic and the information-theoretic bookkeeping of the symmetry breaking agree exactly, with no correction term from the type III structure — a point made independently in Lemma 5.10, where the relative entropy is shown to be purely central.

6. THE CONTROL EXPERIMENT: RESTORING THE ADDITIVE PART

To confirm that Lemma 1.1 is the cause of §§2–5, we restore the missing ingredient.

Remark 6.1 (The additive group is $\mathcal{O}_K$). The ring of S -integers $\mathcal{O}_{K,S}$ inverts the primes of S , so its elements have poles at S and it is not contained in $\widehat{\mathcal{O}}_S$. The correct additive part is $\mathcal{O}_K$, dense in $\widehat{\mathcal{O}}_S$ by the Chinese remainder theorem. We let J_S act on $\widehat{\mathcal{O}}_S$ through a fixed choice of uniformizers $\pi_{\mathfrak{p}} \in K_{\mathfrak{p}}$, $\mathfrak{a} = \prod \mathfrak{p}^{a_{\mathfrak{p}}}$ acting by $x \mapsto \pi_{\mathfrak{a}} x$ with $(\pi_{\mathfrak{a}} x)_{\mathfrak{p}} = \pi_{\mathfrak{p}}^{a_{\mathfrak{p}}} x_{\mathfrak{p}}$, and set $\mathcal{A}_{K,S}^{\text{aff}} = C(\widehat{\mathcal{O}}_S) \rtimes (\mathcal{O}_K \rtimes J_S)$, the semigroup crossed product by the maps $x \mapsto \pi_{\mathfrak{a}} x + b$. Over $\mathbb{Q}$ the choice $\pi_{\ell} = \ell$ is canonical; in general everything below holds for every choice. (Without the additive part, the choice is absorbed by the G_S -coordinate of Y_S ; with it, the algebra is that of Cuntz–Li [4] and Cuntz–Deninger–Laca [3] localized at S .)

Lemma 6.2 (Finite covers). *For every $\mathfrak{p} \in S$, $\widehat{\mathcal{O}}_S = \bigsqcup_{a \in \mathcal{O}_K/\mathfrak{p}} (a + \mathfrak{p}\widehat{\mathcal{O}}_S)$, a partition into $N_{\mathfrak{p}}$ clopen sets. The isometries $\mu_{(a,\mathfrak{p})}$ therefore have mutually orthogonal ranges and satisfy the Cuntz relation*

$$(6.1) \quad \sum_{a \in \mathcal{O}_K/\mathfrak{p}} \mu_{(a,\mathfrak{p})} \mu_{(a,\mathfrak{p})}^* = 1,$$

and the tight spectrum is all of $\widehat{\mathcal{O}}_S$: no collapse occurs, and $\mathcal{A}_{K,S}^{\text{aff}}$ is its own boundary quotient, $\partial \mathcal{A}_{K,S}^{\text{aff}} = \mathcal{A}_{K,S}^{\text{aff}}$.

Proof. $\mathfrak{p}\widehat{\mathcal{O}}_S$ is open of index $N\mathfrak{p}$ and $\mathcal{O}_K$ surjects onto $\widehat{\mathcal{O}}_S/\mathfrak{p}\widehat{\mathcal{O}}_S$ by density. The cover is finite, so the tight relations of Exel [5] impose equality rather than the nested inequality of Lemma 1.1; and (6.1) already holds in $\mathcal{A}_{K,S}^{\text{aff}}$, since the Stacey covariance gives $\mu_{(a,\mathfrak{p})}\mu_{(a,\mathfrak{p})}^* = \mathbf{1}_{a+\mathfrak{p}\widehat{\mathcal{O}}_S}$. $\square$

Theorem 6.3 (Kirchberg boundary). *The action of $\mathcal{O}_K \rtimes J_S$ on $\widehat{\mathcal{O}}_S$ is minimal and topologically free, and $\partial\mathcal{A}_{K,S}^{\text{aff}}$ is a unital separable nuclear simple purely infinite C^* -algebra satisfying the UCT.*

Proof. Minimality: $\mathcal{O}_K$ is dense and acts by translations, so every orbit is dense. Topological freeness: a nontrivial $(b, \mathfrak{a})$ with $\mathfrak{a} = 1$ is a nontrivial translation of a group, hence fixed-point free. For $\mathfrak{a} \neq 1$ the fixed points satisfy $(1 - \pi_{\mathfrak{a}})x = b$ componentwise: at $\mathfrak{p} \in \text{supp } \mathfrak{a}$ the element $1 - \pi_{\mathfrak{p}}^{a_{\mathfrak{p}}}$ is a unit, so $x_{\mathfrak{p}}$ is determined; at $\mathfrak{p} \notin \text{supp } \mathfrak{a}$ the equation reads $0 = b$ in $\mathcal{O}_{\mathfrak{p}}$, which forces $b = 0$. So the fixed-point set is empty, or a single point, or (for $b = 0$) the set $\{x : x_{\mathfrak{p}} = 0 \ \forall \mathfrak{p} \in \text{supp } \mathfrak{a}\}$; in every case it is closed with empty interior. The union over the countable semigroup is meagre, hence has empty interior. The groupoid of germs is étale, Hausdorff and amenable, being that of a partial action of the amenable group $K \rtimes I_S$; minimality and topological freeness give simplicity [2]. It is locally contracting: a basic clopen set $V = \{x : x_{\mathfrak{q}} \in a_{\mathfrak{q}} + \mathfrak{q}^{n_{\mathfrak{q}}}\mathcal{O}_{\mathfrak{q}}, \ \mathfrak{q} \in F\}$ is mapped by $x \mapsto \pi_{\mathfrak{p}}x + c$, $\mathfrak{p} \in F$ and $c \in \mathcal{O}_K$ chosen by the Chinese remainder theorem with $c \equiv (1 - \pi_{\mathfrak{p}})a_{\mathfrak{p}} \pmod{\mathfrak{p}^{n_{\mathfrak{p}}}}$ and $c \equiv 0 \pmod{\mathfrak{q}^{n_{\mathfrak{q}}}}$ for $\mathfrak{q} \neq \mathfrak{p}$, onto a clopen proper subset of V . Hence $\mathcal{A}_{K,S}^{\text{aff}}$ is purely infinite by Anantharaman-Delaroche [1]. Nuclearity and the UCT follow from amenability. $\square$

Theorem 6.4 (The unit class says little even here). *In $K_0(\partial\mathcal{A}_{K,S}^{\text{aff}})$ one has $(N\mathfrak{p} - 1)[1] = 0$ for every $\mathfrak{p} \in S$, so the order of $[1]$ divides $d_S = \gcd_{\mathfrak{p} \in S}(N\mathfrak{p} - 1)$. In particular $[1] = 0$ whenever $d_S = 1$, which holds as soon as S contains a prime of norm 2, or two primes with $\gcd(N\mathfrak{p} - 1, N\mathfrak{q} - 1) = 1$.*

Proof. Each $\mu_{(a,\mathfrak{p})}$ is an isometry, so $[\mu\mu^*] = [1]$; applying K_0 to (6.1) gives $[1] = N\mathfrak{p}[1]$. $\square$

Proposition 6.5 (The Cuntz relation pins the temperature). *For the dynamics $\sigma_t(\mu_{(b,\mathfrak{a})}) = N\mathfrak{a}^{it}\mu_{(b,\mathfrak{a})}$ on $\mathcal{A}_{K,S}^{\text{aff}}$, a KMS_{β} state exists only for $\beta = 1$.*

Proof. For a KMS_{β} state φ and any of the $N\mathfrak{p}$ isometries in (6.1), the KMS condition gives $\varphi(\mu\mu^*) = \varphi(\mu^*\sigma_{i\beta}(\mu)) = N\mathfrak{p}^{-\beta}\varphi(\mu^*\mu) = N\mathfrak{p}^{-\beta}$; summing (6.1) gives $1 = N\mathfrak{p}^{1-\beta}$, so $\beta = 1$. $\square$

Example 6.6. Over $\mathbb{Q}$: $S = \{3\}$ gives $d_S = 2$; $S = \{3, 5\}$ gives $\gcd(2, 4) = 2$; $S = \{7, 13\}$ gives $\gcd(6, 12) = 6$; but any S containing 2 gives $d_S = 1$ and $[1] = 0$. A single small prime erases the invariant.

Remark 6.7 (Two behaviours of the unit class). Theorems 3.3 and 6.4 exhibit opposite behaviours of $[1]$, and the contrast is instructive.

In the multiplicative boundary, $K_0 = \Lambda^{\text{ev}}(I_S^0)$ is torsion-free and $[1] = h_S^+ e_0$ is a nonzero element of divisibility exactly h_S^+ : the narrow class number is stored in *how divisible* the unit is. In the affine boundary the Cuntz relation forces $[1] = N\mathfrak{p}[1]$, so $[1]$ is a torsion element of order dividing $\gcd_{\mathfrak{p}}(N\mathfrak{p} - 1)$ and is usually zero. The mechanism is already visible for the Cuntz algebra $\mathcal{O}_n$, where $K_0 = \mathbb{Z}/(n - 1)$ is generated by $[1]$; taking several primes replaces $n - 1$ by a greatest common divisor, and a single prime of norm 2 collapses it.

So the two systems store arithmetic in $[1]$ in incompatible ways, and only the multiplicative one does so usefully. This is a rare instance in which the nested range projections of Lemma 1.1 are an advantage: they leave K_0 torsion-free, and a torsion-free group can record an integer by divisibility, which a group killed by $\gcd_{\mathfrak{p}}(N\mathfrak{p} - 1)$ cannot. We have not computed the ambient groups $K_*(\partial\mathcal{A}_{K,S}^{\text{aff}})$ and make no claim about them beyond Theorem 6.4.

7. ASSESSMENT

Feature	$\mathcal{A}_{K,S}$ (multiplicative)	$\mathcal{A}_{K,S}^{\text{aff}}$ (affine)
isometries per $\mathfrak{p}$	1	$N\mathfrak{p}$
range projections	nested	partitioning
tight relation	$\mu_{\mathfrak{p}}$ unitary	Cuntz
boundary space	Cl_S^+ , finite	$\widehat{\mathcal{O}}_S$, a Cantor group
boundary algebra	$M_{h_S^+}(C^*(I_S^0))$, type I	Kirchberg (the algebra itself)
trace on the boundary	faithful	none
[1] records	h_S^+	order $\mid \gcd(N\mathfrak{p} - 1)$

Everything in the left column follows from Lemma 1.1, and the right column confirms by contrast that the additive cosets are the cause. The same dichotomy governs the admissible inverse temperatures — the Cuntz relation forces $\beta = 1$ on the affine system (Proposition 6.5), whereas the multiplicative system has KMS_β states for every $\beta > 0$ [21, Thm. 3.6] — and the free variant treated in Paper V of the series, where orthogonal ranges again restrict β .

Four questions remain. We have computed $K_*(\mathcal{A}_{K,S})$ only for a single prime, and $K_*(\mathcal{A}_{K,S}^{\text{aff}})$ not at all; by Theorem 6.3 the triple $(K_0, [1], K_1)$ is a complete invariant of the affine algebra and by Theorem 6.4 the middle entry is usually trivial, so the content lies in the groups. When $\mathcal{O}_{K,+}^\times$ is infinite, whether some proper stratum carries nontrivial isotropy (Remark 2.3) — a Leopoldt-type closure question — and hence contributes further ideals of the kind seen in Theorem 2.5(2), is open. Whether the infinite index of Theorem 5.3 persists at the non-symmetric extremal states (Remark 5.4), and whether these inclusions are discrete in the sense of Izumi–Longo–Popa [8] or admit a continuous decomposition matching the diffuse relative commutant, is open. Finally, the lattice of intermediate subfactors [20] of the symmetry-breaking inclusion is the partition lattice of $\widehat{\Xi}_\beta$ (Theorem 5.8), which remembers Ξ_β only as a finite set; it distinguishes neither K nor S nor the thresholds, and the question of which invariant does is taken up in Paper III of the series.

It is worth saying plainly why the blindness recorded in Remark 4.5 is not a computational accident. By Propositions 4.2 and 4.3 the passage from $C(Y_S)$ to $\mathcal{A}_{K,S}$ is, on the image of $K^0(Y_S)$ in K_0 , the passage to the coinvariants of α_* , and a functional survives that quotient exactly when it is α_* -invariant. The arithmetically meaningful functionals are the KMS_β measures, which by [21, Prop. 2.8] are eigenvectors, $\omega \circ \alpha_{\mathfrak{p}*} = N\mathfrak{p}^{-\beta}\omega$, with eigenvalue $\neq 1$ for $\beta \neq 0$. Passing to the coinvariants therefore *annihilates precisely the covariant vectors that carry the arithmetic*, by construction rather than by accident. The same mechanism in its measure-theoretic form is, we expect, the reason Ξ_β is an invariant of a measure class rather than of a C^* -algebra: any functor that forgets α_* , or that forces it to act as the identity, loses it.

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